Lecture 9: Trees, Cutsets and Fundamental Loops

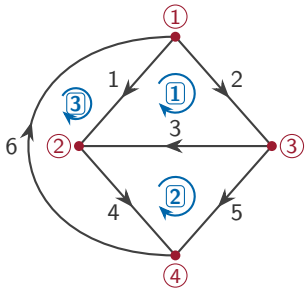
EE201 – Circuit Theory I

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Middle East Technical University

current voltage, cutsets, loops tree nodes

Review: two matrices for one graph



$n = 4, \quad b = 6$

Incidence matrix (rows: nodes ①, ②, ③)

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Mesh matrix (rows: meshes 1, 2, 3)

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

$n - 1 = 3$ independent KCL equations, $\quad b - n + 1 = 3$ independent KVL equations

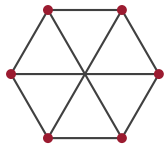
Every row of A is orthogonal to every row of M: $\quad AM^T = \mathbf{0}$

Review: node voltages and mesh currents

	node voltages	**mesh currents**
works for	any connected graph	planar, nonseparable
number of unknowns	$n-1$	$b-n+1$
holds automatically	KVL: $\mathbf{v} = A^T\mathbf{e}$	KCL: $\mathbf{i} = M^T\mathbf{i}_M$
left to write	KCL: $A\mathbf{i} = \mathbf{0}$	KVL: $M\mathbf{v} = \mathbf{0}$

Tellegen: $\mathbf{i}^T\mathbf{v} = 0$

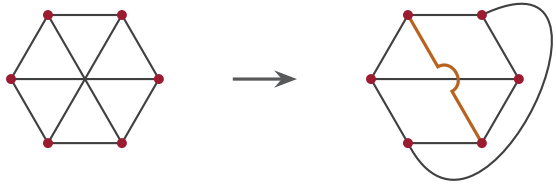
Nonplanar circuits



crossings at the centre

Can this graph be redrawn without crossings?

Nonplanar circuits

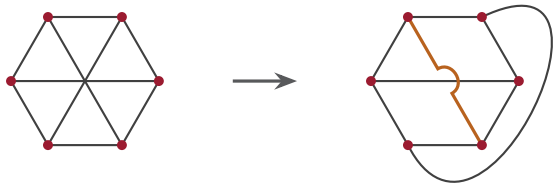


crossings at the centre

jumper wire: still one hop

No: at least one hop remains ⇒ no meshes, no mesh matrix.

Nonplanar circuits



crossings at the centre

jumper wire: still one hop

No: at least one hop remains $\Rightarrow$ no meshes, no mesh matrix.

Question

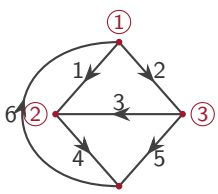
Without meshes, how do we find $n-1$ independent KCL and $b-n+1$ independent KVL equations?

Today: a method that needs no planar drawing and works for any connected graph.

Trees

Tree

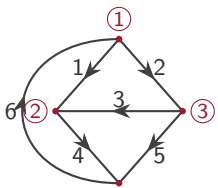
A subgraph T of a connected graph G is a tree if it contains all the nodes of G, is connected, and contains no loops.



Trees

Tree

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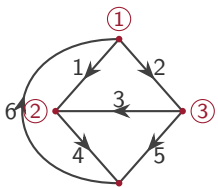


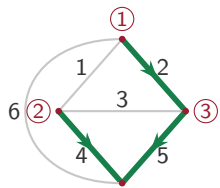
graph G $\{2, 4, 5\}$**: tree?**

Trees

Tree

A subgraph T of a connected graph G is a tree if it contains all the nodes of G, is connected, and contains no loops.

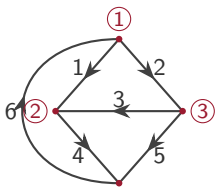




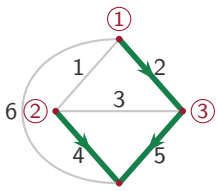
Trees

Tree

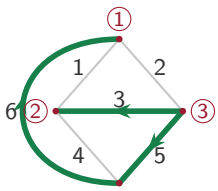
A subgraph T of a connected graph G is a tree if it contains all the nodes of G, is connected, and contains no loops.



graph G



$\{2, 4, 5\}$: **tree**

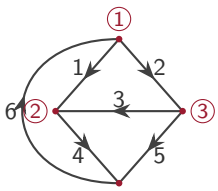


$\{3, 5, 6\}$: **tree?**

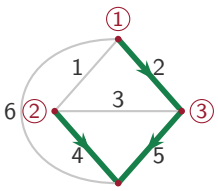
Trees

Tree

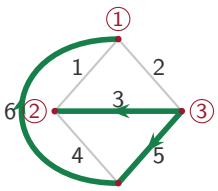
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graph G



$\{2, 4, 5\}$: **tree**

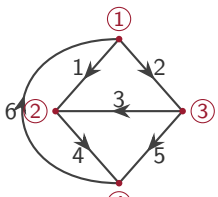


$\{3, 5, 6\}$: **tree**

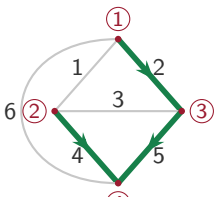
Trees

Tree

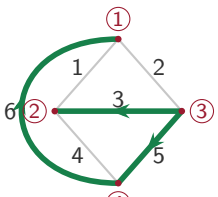
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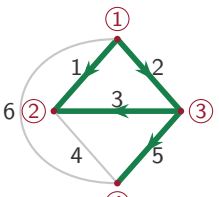
graph G



$\{2, 4, 5\}$: **tree**



$\{3, 5, 6\}$: **tree**

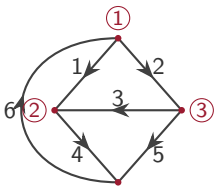


$\{1, 2, 3, 5\}$: **tree?**

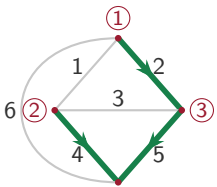
Trees

Tree

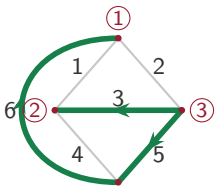
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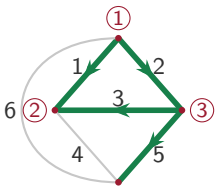
graph G



$\{2, 4, 5\}$: **tree**



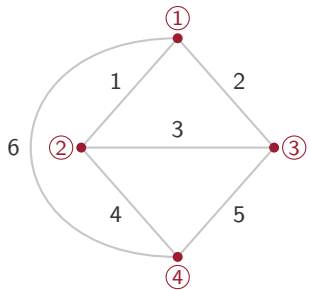
$\{3, 5, 6\}$: **tree**



$\{1, 2, 3, 5\}$: **not a tree**

Branches 1, 2, 3 form a loop. A graph has many trees.

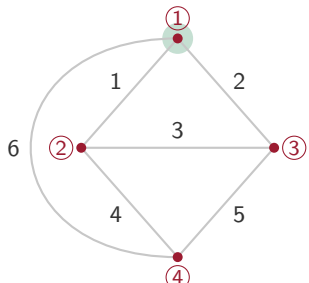
How many branches does a tree have?



$n = 4$ nodes, $b = 6$ branches

How many branches does a tree have?

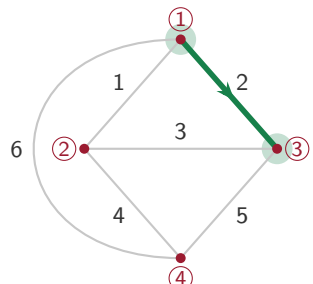
- Start from one node; add tree branches one at a time.



start at node ①

How many branches does a tree have?

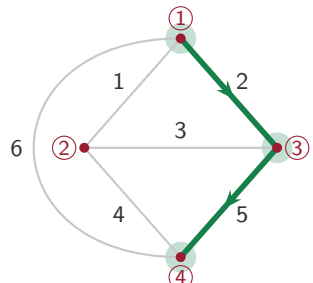
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 2 reaches node ③

How many branches does a tree have?

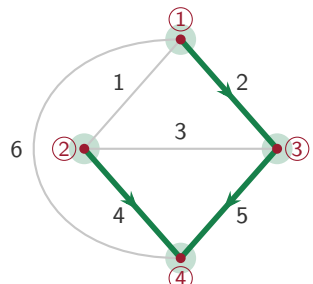
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 5 reaches node ④

How many branches does a tree have?

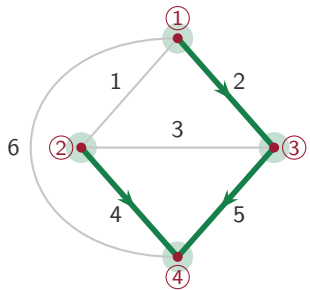
- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.



branch 4 reaches node ②

How many branches does a tree have?

- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.
- Each branch adds one node. All n nodes are reached after $n-1$ branches.

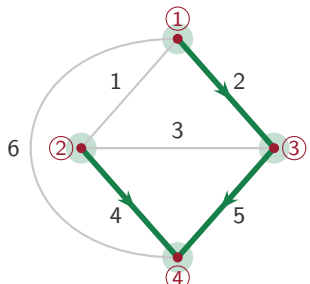


4 nodes, 3 tree branches: $n-1=3$

How many branches does a tree have?

- Start from one node; add tree branches one at a time.
- Each new branch must reach a new node — otherwise it closes a loop.
- Each branch adds one node. All n nodes are reached after $n-1$ branches.

The other $b-n+1$ branches are left out.



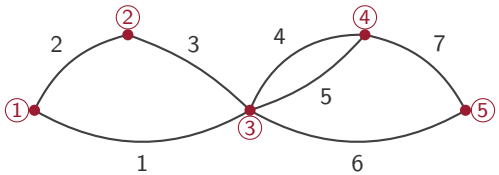
left out: 1, 3, 6 $(b-n+1=3)$

Cutsets

Cutset

A set C of branches of a connected graph is a cutset if

1. removing all of C leaves the graph unconnected, and
2. removing all of C except any one of them leaves it connected.

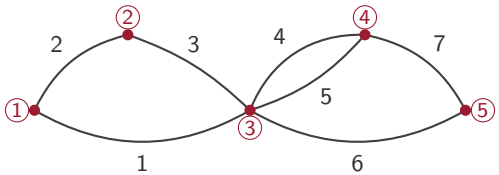


Cutsets

Cutset

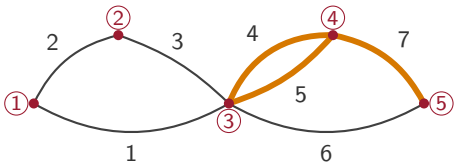
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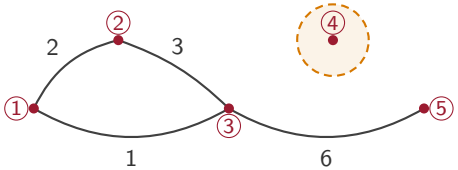
In words: removing the set cuts the graph into two pieces, and every branch in the set is needed for that.

Example: cutsets



$$C_1 = \{4, 5, 7\}$$

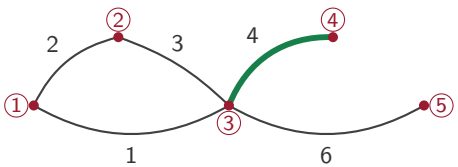
Example: cutsets



$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

Example: cutsets



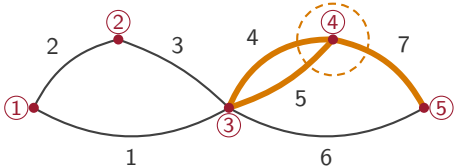
$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

(2) Put back one, e.g. 4: node ④ is reached again. Same for 5 or 7. ✓

$\Rightarrow$ C_1 is a cutset.

Example: cutsets



$C_1 = \{4, 5, 7\}$

(1) Remove all three: node ④ is cut off. ✓

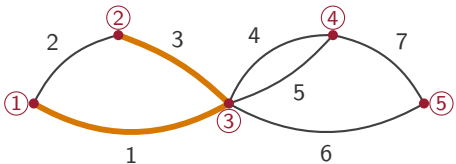
(2) Put back one, e.g. 4: node ④ is reached again. Same for 5 or 7. ✓

$\Rightarrow$ C_1 is a cutset.

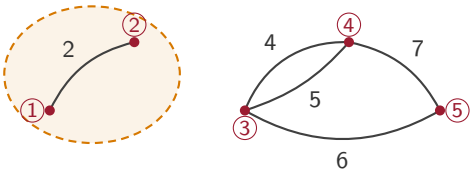
All branches back: the closed surface around node ④ crosses exactly 4, 5, 7.

Example: cutsets

$$C_2 = \{1, 3\}$$



Example: cutsets



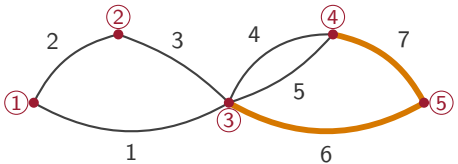
$C_2 = \{1, 3\}$

(1) Remove both: nodes ①, ② are cut off. ✓

(2) Put back 1 or 3: connected again. ✓

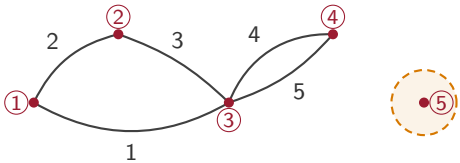
$\Rightarrow$ C_2 is a cutset.

Example: cutsets



$$C_3 = \{6, 7\}$$

Example: cutsets



$C_3 = \{6, 7\}$

(1) Remove both: node ⑤ is cut off. ✓

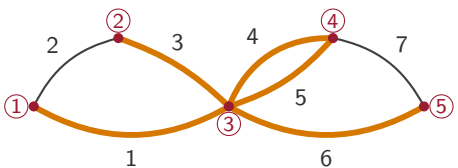
(2) Put back 6 or 7: connected again. ✓

$\Rightarrow$ C_3 is a cutset.

Cutsets can have different sizes: C_1 has 3 branches, C_2 and C_3 have 2.

Example: not every set is a cutset

$C = \{1, 3, 4, 5, 6\}$: a cutset?



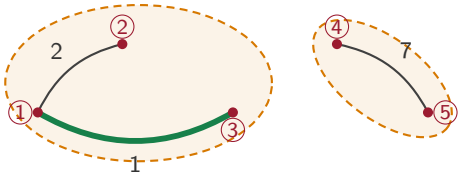
Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

Example: not every set is a cutset



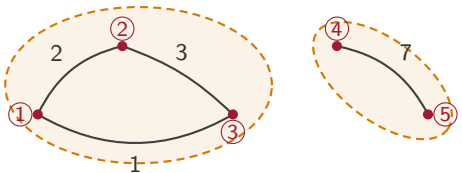
$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

(2) Put back 1: still unconnected. ×

$\Rightarrow$ not a cutset: some branches are not needed.

Example: not every set is a cutset



$C = \{1, 3, 4, 5, 6\}$: a cutset?

(1) Remove all five: three pieces. ✓

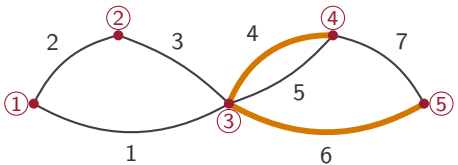
(2) Put back 1: still unconnected. ×

$\Rightarrow$ not a cutset: some branches are not needed.

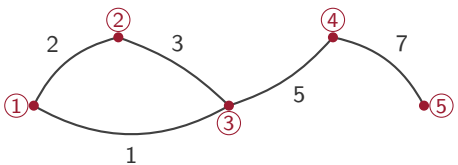
Branches 1, 3 are extra: $\{4, 5, 6\}$ alone is a cutset.

Example: not every set is a cutset

$C = \{4, 6\}$: a cutset?



Example: not every set is a cutset

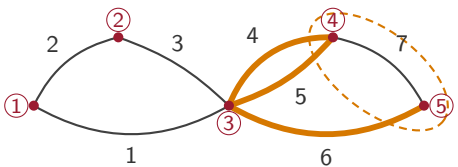


$C = \{4, 6\}$: a cutset?

(1) Remove both: still one piece — branch 5 still joins ③ and ④. ×

$\Rightarrow$ not a cutset: a branch is missing.

Example: not every set is a cutset



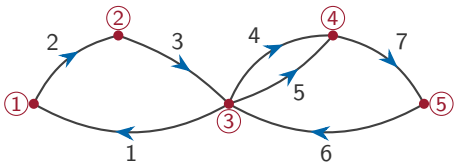
$C = \{4, 6\}$: a cutset?

(1) Remove both: still one piece — branch 5 still joins ③ and ④. ×

$\Rightarrow$ not a cutset: a branch is missing.

Branches 4 and 5 both join ③ and ④ (parallel): a surface that crosses one crosses the other. Here the surface gives $\{4, 5, 6\}$.

KCL on a cutset



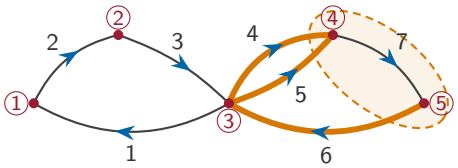
branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Branch currents in the arrow directions.

They satisfy KCL at every node, e.g. node ③:

$$i_1 + i_4 + i_5 - i_3 - i_6 = 2+1+2-2-3 = 0.$$

KCL on a cutset



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Cutset $\{4, 5, 6\}$: surface around ④, ⑤

leaving: i_6; entering: i_4, i_5

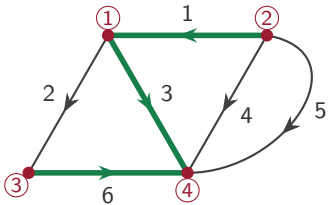
$$i_6 - i_4 - i_5 = 3 - 1 - 2 = 0 \checkmark$$

KCL at ④: $i_7 - i_4 - i_5 = 0$
KCL at ⑤: $i_6 - i_7 = 0$

Add them: i_7 is inside the surface and cancels.

KCL on a cutset: the cutset currents add up to zero
(+ leaving the surface, − entering).

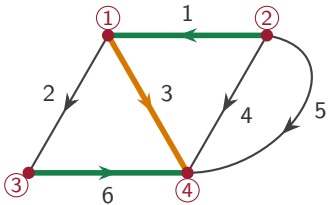
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.

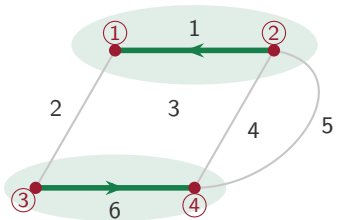
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.
2. Pick one tree branch, e.g. 3.

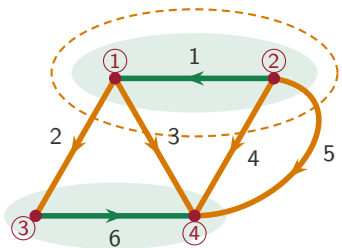
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.
2. Pick one tree branch, e.g. 3.
3. Remove it: the tree falls into two parts, $\{①, ②\}$ and $\{③, ④\}$.

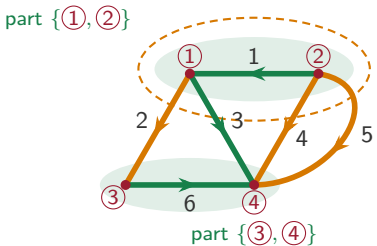
Fundamental cutsets



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

1. Fix a tree (green). The other branches form the co-tree.
2. Pick one tree branch, e.g. 3.
3. Remove it: the tree falls into two parts, $\{①, ②\}$ and $\{③, ④\}$.
4. The branches joining the two parts form a cutset: $\{3, 2, 4, 5\}$. It is the fundamental cutset of 3.

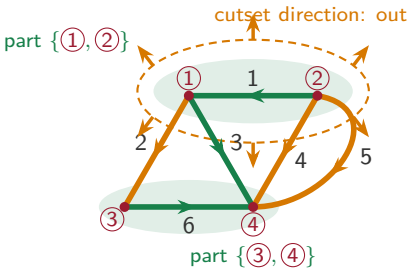
Fundamental cutset: one tree branch, one direction



Cutset of 3: $\{3, 2, 4, 5\}$

One tree branch. Only 3 is a tree branch; 2, 4, 5 are co-tree branches. The other tree branches (1, 6) lie inside one part.

Fundamental cutset: one tree branch, one direction

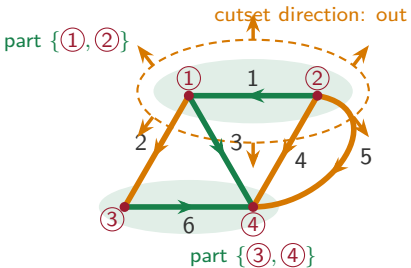


Cutset of 3: $\{3, 2, 4, 5\}$

One tree branch. Only 3 is a tree branch; 2, 4, 5 are co-tree branches. The other tree branches (1, 6) lie inside one part.

Direction. Branch 3 leaves the surface, from part $\{①, ②\}$ to part $\{③, ④\}$: the cutset points out.

Fundamental cutset: one tree branch, one direction



Cutset of 3: $\{3, 2, 4, 5\}$

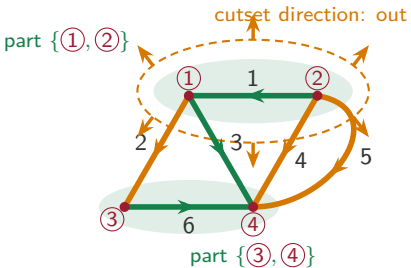
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KCL. Branch k carries current i_k along its arrow. KCL on the surface: $+$ leaving, $-$ entering. Branches 2, 4, 5 leave too:

$$i_3 + i_2 + i_4 + i_5 = 0.$$

Fundamental cutset: one tree branch, one direction



Fundamental cutset

- the tree branch + the co-tree branches joining the two parts
- only one tree branch in it
- direction: that of the tree branch

Cutset of 3: $\{3, 2, 4, 5\}$

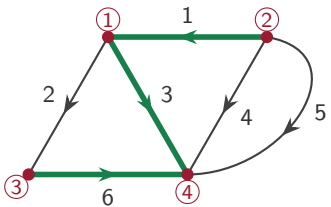
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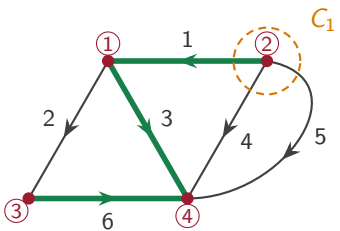
Example: fundamental cutsets of G



For each tree branch: its fundamental cutset, and KCL on that cutset (branch currents i_k along the arrows; $+$ if a branch crosses the surface the same way as the tree branch, $-$ if not).

tree br.	cutset	KCL on it

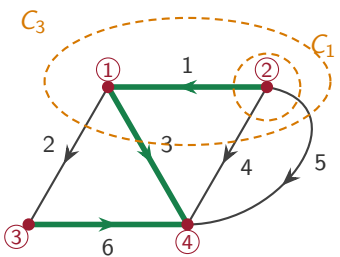
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tree br.	cutset	KCL on it
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$

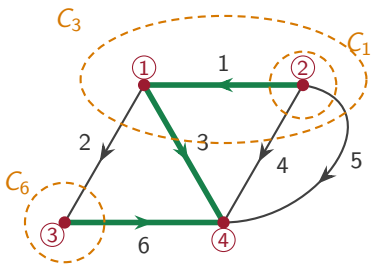
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tree br.	cutset	KCL on it
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$
3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$

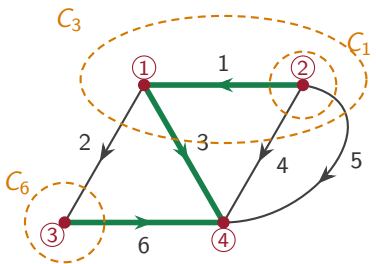
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tree br.	cutset	KCL on it
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$
3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$
6	$\{6, 2\}$	$i_6 - i_2 = 0$

Example: fundamental cutsets of G

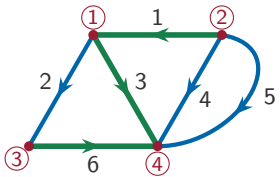


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$n - 1 = 3$ KCL equations, independent: each has its own tree branch current.

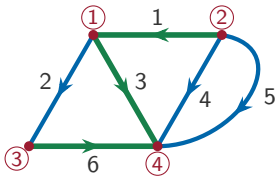
Co-tree currents are enough



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

Question. We know only the co-tree currents i_2, i_4, i_5. Can we find all the other currents?

Co-tree currents are enough



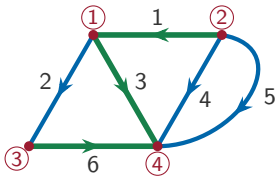
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Each fundamental cutset has one tree branch. Solve its KCL for that current:

$$i_1 = -i_4 - i_5, \quad i_3 = -i_2 - i_4 - i_5, \quad i_6 = i_2$$

Co-tree currents are enough



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

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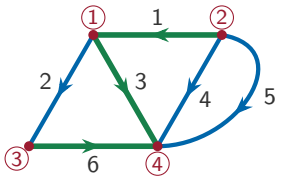
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$$i_1 = -i_4 - i_5, \quad i_3 = -i_2 - i_4 - i_5, \quad i_6 = i_2$$

For example $i_2 = 1$, $i_4 = 1$, $i_5 = 2$ A gives

$$i_1 = -3, \quad i_3 = -4, \quad i_6 = 1 \text{ A}.$$

Co-tree currents are enough



tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

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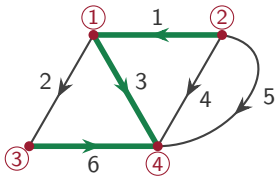
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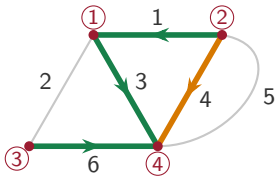
Yes. So the unknown currents reduce from b to the $b - n + 1$ co-tree currents, and KCL holds automatically — the same role as mesh currents.

Fundamental loops



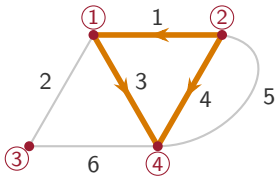
1. Fix a tree (green): $\{1, 3, 6\}$. Co-tree: $\{2, 4, 5\}$.

Fundamental loops



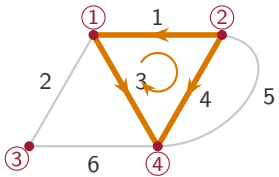
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Fundamental loops



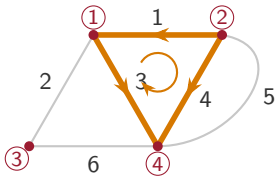
1. Fix a tree (green): $\{1, 3, 6\}$. Co-tree: $\{2, 4, 5\}$.
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3. Add it to the tree: it closes exactly one loop, branch 4 plus the tree path between its ends (3 and 1).

Fundamental loops



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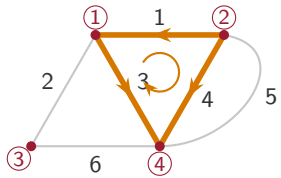
Fundamental loops



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5. KVL around it: $+v_k$ along an arrow, $-v_k$ against:

$$v_4 - v_3 - v_1 = 0.$$

Fundamental loops



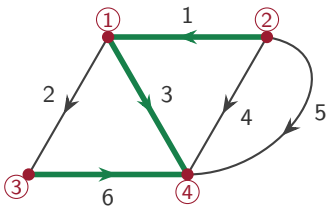
Fundamental loop

- the co-tree branch + the tree path between its ends
- only one co-tree branch in it
- direction: that of the co-tree branch

1. Fix a tree (green): $\{1, 3, 6\}$. Co-tree: $\{2, 4, 5\}$.
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Example: fundamental loops of G

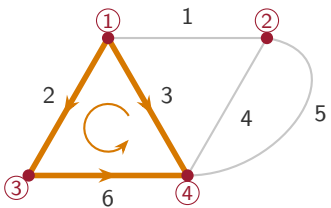


tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

For each co-tree branch: its fundamental loop, and KVL around it (walk in the direction of the co-tree branch; $+v_k$ along an arrow, $-v_k$ against).

co-tree br.	loop	KVL around it

Example: fundamental loops of G

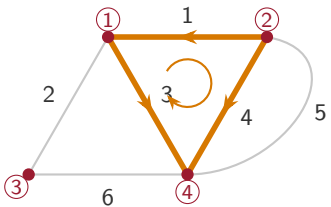


tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

For each co-tree branch: its fundamental loop, and KVL around it (walk in the direction of the co-tree branch; $+v_k$ along an arrow, $-v_k$ against).

co-tree br.	loop	KVL around it
2	$\{2, 6, 3\}$	$v_2 + v_6 - v_3 = 0$

Example: fundamental loops of G

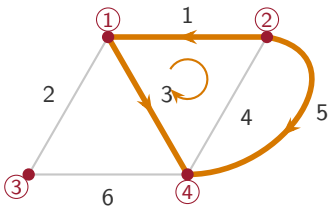


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Example: fundamental loops of G

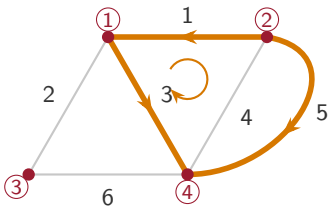


tree $\{1, 3, 6\}$, co-tree $\{2, 4, 5\}$

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co-tree br.	loop	KVL around it
2	$\{2, 6, 3\}$	$v_2 + v_6 - v_3 = 0$
4	$\{4, 3, 1\}$	$v_4 - v_3 - v_1 = 0$
5	$\{5, 3, 1\}$	$v_5 - v_3 - v_1 = 0$

Example: fundamental loops of G



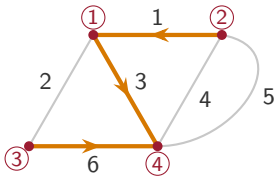
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$b - n + 1 = 3$ KVL equations, independent: each has its own co-tree branch voltage.

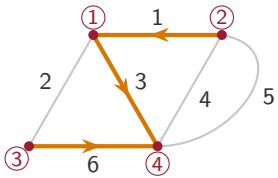
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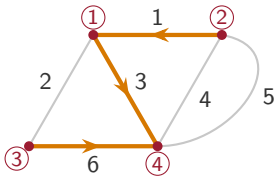
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$$v_2 = v_3 - v_6, \quad v_4 = v_1 + v_3, \quad v_5 = v_1 + v_3$$

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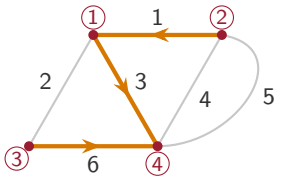
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For example $v_1 = 2$, $v_3 = 3$, $v_6 = 1$ V gives

$$v_2 = 2, \quad v_4 = 5, \quad v_5 = 5 \text{ V}.$$

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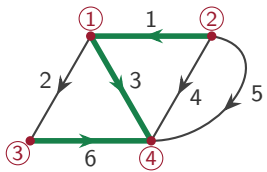
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Yes. So the unknown voltages reduce from b to the $n - 1$ tree voltages, and KVL holds automatically — the same role as node voltages.

Matrix form: fundamental cutset matrix Q



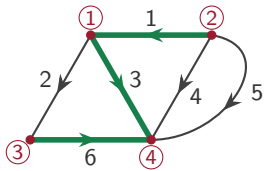
tree $\{1, 3, 6\}$

Collect the fundamental cutset KCL equations in a matrix:

one row per tree branch, one column per branch;
$+1$ / -1: branch in the cutset, same / opposite direction;
0: not in it.

$$Q = \begin{array}{c} \\ 1 \\ 3 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

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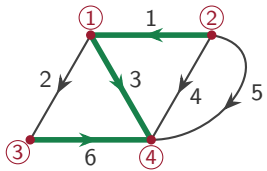
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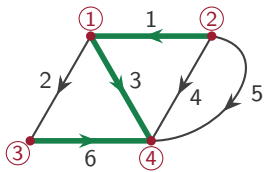
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KCL: $\mathbf{Q i = 0}$ (like $\mathbf{A i = 0}$)

Tree columns 1, 3, 6 form an identity: each row has its own tree branch, so the rows are independent.

Matrix form: fundamental loop matrix B



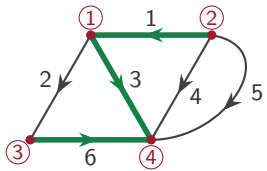
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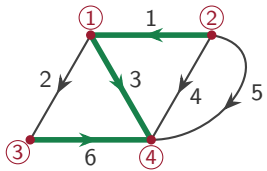
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Co-tree columns 2, 4, 5 form an identity: each row has its own co-tree branch, so the rows are independent.

The transposes: currents from B^T, voltages from Q^T

Co-tree currents $\mathbf{i}_C = [i_2 \;\; i_4 \;\; i_5]^T$, tree voltages $\mathbf{v}_T = [v_1 \;\; v_3 \;\; v_6]^T$

$$\mathbf{i} = B^T \mathbf{i}_C : \quad \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} i_2 \\ i_4 \\ i_5 \end{bmatrix}$$

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And the two fit together: $QB^T = \mathbf{0}$, like $AM^T = \mathbf{0}$.

Any connected graph

	how many	gives
tree branches	$n-1$	fundamental cutsets: the KCL equations
co-tree branches	$b-n+1$	fundamental loops: the KVL equations

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No drawing needed: works for nonplanar graphs.

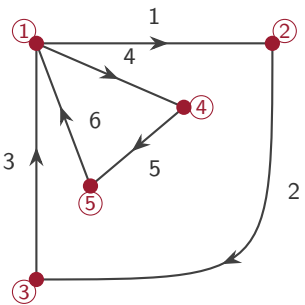
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It also works for separable graphs. Example next.

Example: the separable graph

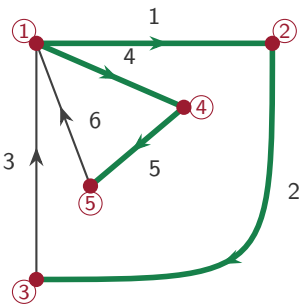


$n = 5, \quad b = 6$

Two loops joined only at node ①.

Separable: the mesh method did not work here.

Example: the separable graph

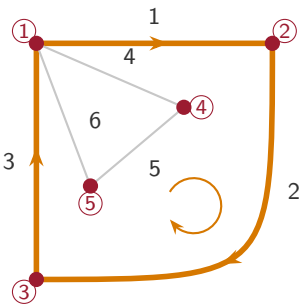


$n = 5, \quad b = 6$

Tree $\{1, 2, 4, 5\}$: $n - 1 = 4$ branches.

Co-tree $\{3, 6\}$: $b - n + 1 = 2$ branches.

Example: the separable graph



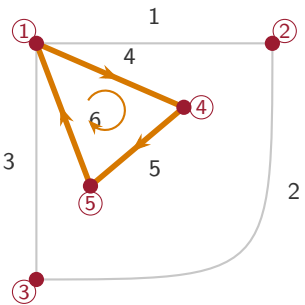
$n = 5, \quad b = 6$

Fundamental loops: KVL around each, walking with the co-tree branch

co-tree br.	loop	KVL
3	$\{3, 1, 2\}$	$v_3 + v_1 + v_2 = 0$

The loop of 3 is the outer loop.

Example: the separable graph



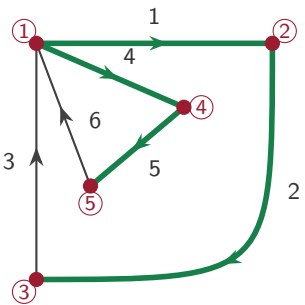
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co-tree br.	loop	KVL
3	$\{3, 1, 2\}$	$v_3 + v_1 + v_2 = 0$
6	$\{6, 4, 5\}$	$v_6 + v_4 + v_5 = 0$

The loop of 6 is the inner triangle.

Example: the separable graph

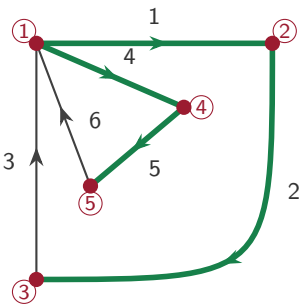


$n = 5, \quad b = 6$

Fundamental cutsets: KCL on each, signs set by the tree branch

tree br.	cutset	KCL
1	$\{1,3\}$	$i_1 - i_3 = 0$
2	$\{2,3\}$	$i_2 - i_3 = 0$
4	$\{4,6\}$	$i_4 - i_6 = 0$
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Example: the separable graph



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So $i_1 = i_2 = i_3$ and $i_4 = i_5 = i_6$:
one current for each loop.

4 KCL + 2 KVL equations, all independent — no problem at node ①.

Looking ahead

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages: $\mathbf{v} = A^T\mathbf{e}$	tree voltages: $\mathbf{v} = Q^T\mathbf{v}_T$
$b-n+1$ currents	mesh currents: $\mathbf{i} = M^T\mathbf{i}_M$	co-tree currents: $\mathbf{i} = B^T\mathbf{i}_C$
works for	A: any; M: planar, nonseparable	any connected graph

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	incidence / mesh	**fundamental cutset / loop**
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works for	A: any; M: planar, nonseparable	any connected graph

These unknowns are what circuit analysis will solve for:
far fewer than the $2b$ branch currents and voltages.

Summary

- **Tree**: all nodes, connected, no loop; $n-1$ branches. The rest: the **co-tree**, $b-n+1$ branches.
- **Cutset**: minimal set of branches that cuts the graph in two. KCL holds on it.
- **Fundamental cutsets** (one per tree branch): $n-1$ independent KCL equations. Knowing the co-tree currents gives all branch currents.
- **Fundamental loops** (one per co-tree branch): $b-n+1$ independent KVL equations. Knowing the tree voltages gives all branch voltages.
- Matrix form: $Q\mathbf{i}=\mathbf{0}$, $B\mathbf{v}=\mathbf{0}$, $\mathbf{i}=B^T\mathbf{i}_C$, $\mathbf{v}=Q^T\mathbf{v}_T$.
- Works for any connected graph — planar or not, separable or not.

Questions?